

Tushar Jindal

Coquitlam, BC ■ tushar.bzp05@gmail.com ■ 236-412-0273
LinkedIn ■ Portfolio ■ GitHub

EDUCATION

University of British Columbia (UBC)

Bachelor of Science in Mathematics (Computational)

Vancouver, BC

Expected May 2027

SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, SQL

Frameworks/Tools: FastAPI, Flask, React, Node.js, LangChain, TensorFlow, Gemini API, spaCy, scikit-learn, ChromaDB, Docker, Git, GitHub Actions

Areas: Machine Learning, NLP, RAG systems, Computer Vision, API design, Data Pipelines, Cloud (AWS)

ACHIEVEMENTS

Hackcamp 2025 (nwPlus): 18-hour hackathon; collaboratively built Code Lantern in a team of 3, focusing on rapid prototyping and AI-powered developer tooling.

CERTIFICATIONS

- **Software Engineer Intern Certificate — HackerRank:** Skills tested in SQL, C++, debugging, and algorithmic reasoning.
- **Supervised ML — Stanford Online:** Regression, classification, regularization, and model evaluation.
- **JavaScript Masterclass — Ultimate Courses:** Scalable JS architecture, async patterns, and performance tuning.

PROJECTS

Code Lantern — FastAPI, Python, Gemini API, JS

Hackcamp 2025

github.com/naomichenruoxi/code-lantern

- Built an **AI-powered architecture analyzer** reconstructing file structure, function relationships, and dependency graphs from uploaded projects.
- Engineered a **regex + AST hybrid parser** across Python/JS/TSX, improving parsing stability and enabling deeper static analysis.
- Delivered REST APIs and a responsive JS frontend with drag-and-drop uploads, interactive views, and explanation panels powered by Gemini 1.5.

Resume Optimizer — Python, FastAPI, spaCy, scikit-learn, React/TS

github.com/Tjindl/resOptimizer

- Built a resume intelligence engine performing **ATS checks, section detection, skill extraction**, and formatting analysis through NLP.
- Implemented spaCy NLP + PyPDF2/python-docx preprocessing for clean text extraction and robust handling of varied résumé layouts.
- Developed a TF-IDF + cosine similarity scoring model with a React dashboard showing relevance scores and improvement suggestions.

UBC Course Assistant — Python, ChromaDB, MiniLM, LangChain

github.com/Tjindl/ubc-course-assistant

- Built a **semantic course chatbot** using MiniLM embeddings + ChromaDB vector search to interpret complex academic queries.
- Added fuzzy lookup, intent routing, and structured RAG templates for clean, well-formatted course recommendations.
- Optimized storage and retrieval pipelines for fast, sub-second responses across thousands of courses.

ASL Recognition — Python, TensorFlow, Flask, React

github.com/Tjindl/asl

- Built a real-time **ASL recognition system** using a TensorFlow CNN trained on curated datasets (99.9% accuracy).
- Implemented a webcam inference pipeline with preprocessing, batching, and Flask-based model serving for responsive predictions.
- Built a React UI for live gesture inference, uploads, and interactive model testing.

EXPERIENCE

Tutor — Self-Employed

Dec 2022 – Present

- Teach math and programming with focus on conceptual reasoning, algorithms, and problem-solving.
- Create individualized learning plans and project-based lessons across Python, Java, algebra, and calculus.
- Support students with debugging, exam preparation, mistake analysis, and structured long-term feedback cycles.
- Mentor students on writing cleaner code and organizing project workflows effectively.